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DATA CENTRE INFRASTRUCTURE

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Abstract

Ltd. (HDS Nepal), which was founded in Section A. This section illustrates the choice of enterprise grade networking devices such as switches, routers, firewalls, servers and storage systems that are consistent with the topology of Spine-Leaf network and SDN Overlay architecture. Every hardware recommendation will be backed by technical specifications, market availability and description in accordance to the business needs of HDS Nepal that include scalability, performance, security, and cost-effectiveness. Comparison between types of servers (rack, blade, tower), and storage (DAS, NAS, SAN) are compared to justify decision-making. Also the floor plan is proposed to meet the requirements of TIA-942, which further characterizes important zones like Entrance Room (ER), Main Distribution Area (MDA), Horizontal Distribution Area (HDA), Zone Distribution Area (ZDA), and Equipment Distribution Area (EDA). This part will turn the strategic framework provided in Section A into a feasible and workable infrastructure plan that will respond to the domestic operations issue in Nepal and aligns HDS Nepal to grow sustainably in the competitive digital services industry.

1. Introduction

Himalayan Digital Services Pvt. Ltd. (HDS Nepal) is an IT service provider headquartered in Kathmandu and was founded in 2018 to provide enterprise solutions with cloud computing, managed hosting, SaaS, and digital infrastructure consulting services to more than 150 enterprise clients in the banking, finance, e-commerce, education, and government fields. The key strategic move to move out of a co-located facility to a special Tier III data centre has a way of curbing major constraints in the aspects of scaling, security management, and the ability to tailor services.

Section A defined the underline design parameters: Tier III design based on 99.982% availability, Spine-Leaf network architecture to optimize east-west traffic, SDN Overlay architecture based on VMware NSX to provide auto-provisioning and micro-segmentation, and full power and cooling redundancy. Section B expounds on this by defining and supporting the real IT hardware pieces that have to be implemented.

The following two important deliverables will be discussed in this section:

- IT Hardware Recommendation: Glassus will be selected based on the existing market solutions of switches, routers, firewalls, servers, and storage to support HDS Nepal needs and requirements as per Nepal operations costs and any additional scalling needs.
- TIA-942 Conformant Floor Plan A detailed data centre floor plan with standard areas such as Entrance Room (ER), Main Distribution Area (MDA), Horizontal Distribution Area (HDA), Zone Distribution Area (ZDA) and Equipment Distribution Area (EDA), provision of optimum utilization of space, managing cables, and efficient airflow and compliance to security.

All these elements and design architecture allow HDS Nepal to meet its goals of high availability, maximum performance, security concerns, low energy consumption, scaling, and confronting not only the challenges faced by Nepal such as seismic activity, unstable power infrastructure, and human resource shortages but also other challenges in the country.

2. Data Centre IT Components

2.1 Switches

A network switch is an essential networking solution which can be used in a data centre to connect networking devices and direct the flow of the data packet according to the MAC addresses to create an efficient communication between the servers, storage and network infrastructure (GeeksforGeeks, 2025). In the Spine-Leaf architecture used by HDS Nepal, switches are used in different functions:

- Spine Switches: Spine switches offer connectivity over a spine that is high speed, and they connect all switches in a leaf-to-leaf topology.
- Leaf Switches This type of switch bridges servers (Top-of-Rack) and offers access layer features.
- Management Switches: Full Out-of-band management networks switches.

2.1.1 Selection of Spine Switch

Figure 1: Arista 7050X3-96YC12 Modular Spine Switch

Specifications:

Table 1

Feature	Specification
Ports	96 x 25/50G, 12 x 100G QSFP28 uplinks
Switching Capacity	6.0 Tbps (non-blocking)
Latency	600 ns line-rate latency
Redundancy	Dual hot-swappable PSU and fans
Management	Arista EOS, REST API, SNMP, Streaming Telemetry

Features:

- **High Scalability:** Because of modular uplinks, is scalable to the requirements of future growth.
- **Low Latency:** Less than microsecond latency used in east-west traffic of virtualized environments.
- **Arista EOS:** This is a programmable, automated network control, open APIs.
- **Telemetry:** Performance and congestion monitoring in real time.
- **Redundancy:** The power supplies and fans and are dual and hot-swappable in order to support Tier III concurrent maintainability.

Justification for Selection

The spine switch to be used in HDS Nepal is the Arista 7050X3-96YC12 as it has the capacity to achieve high performance to allow low latency traffic which is used in virtualized workload and real time financial applications. It has a modular structure that enables it to scale with incremental capacity in ports as well as bandwidth without interrupting the operation. Moreover, the switch is SDN-ready, and Arista EOS offers automation, programmability, and the smooth integration with SDN overlay networks. The dual power supplies and fans provide the required redundancy to achieve the Tier III requirements, where concurrent maintenance can be done without affecting service. In addition, the switch is also future-proofed with 100G uplinks, which means that it can support the growing bandwidth needs in the future.

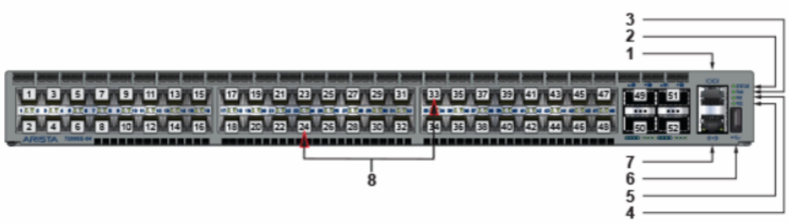
2.1.2 Selection of Leaf Switch (Top-of-Rack)

Arista 7280R Series Leaf Switch

Figure 2: Arista 7280R Series Leaf Switch

Table 2

Feature	Specification
Ports	48 x 25G server ports, 6 x 100G uplinks
Latency	< 600 ns line-rate latency
Hardware Offload	VXLAN/EVPN offload
Management	EOS, REST API, SNMP
Redundancy	Hot-swappable PSU and fans



1	Console port (serial)	4	Power supply 1 status LED	7	Ethernet Management port
2	System status LED	5	Power supply 2 status LED	8	Port numbers
3	Fan status LE	6	USB port		

Figure 3: Font Side of Arista 7280R Series

Key Features:

- **Dense Connections:** there will be 48 server ports per rack and 6 uplinks to spine switch.
- **VXLAN Offload:** The support of VXLAN/EVPN hardware support guarantees zero performance cost.

- **State of the Art Telemetry:** Displays congestion and traffic in real-time.
- **Nano time Accuracy:** PTP High-latency and financial use.
- **Redundant Architecture:** The by-product of concurrent maintainability is through redundant architecture.

Figure 4: Rear to Front and Front to Rear

Justification:

Arista 7280R has been chosen to be used in HDS Nepal because the architecture and operational specifications of the data center highly concur with them. Its architecture is ideal to be a Top-of-Rack switch to effectively interconnect servers with spine switches without bottlenecks forming in the network. Hardware offload using VXLAN/EVPN ensures sub-microsecond latency on the workloads that are critical. The switch is capable of supporting 48 servers per rack without oversubscription of switches in dense operational environments and enables the network to be easily scaled in line with the amplified demand. It is easily compatible with the SDN overlay architecture, allowing more advanced functions, including micro-segmentation and multi-tenant isolation. Streaming telemetry is also used to improve operational efficiency, making it easier to carry out troubleshooting in the environments with low networking expertise. Moreover, the Arista 7280R is also Tier III compliant and supports redundant and hot-swappable components, so maintenance and upgrades can be performed with no downtime.

2.1.3 Selection of Management Switch

Arista 720XP Series (C720XP-48)

Figure 5: Arista 720XP Management Switch

Table 3

Feature	Specification
Ports	48 x Gigabit Ethernet, 4 x 10G SFP+ uplinks
Switching Capacity	176 Gbps
PoE	Supports POE+ up to 740W
Management	CLI, EOS, SNMP, REST API
Redundancy	Modular power supplies and fans

Key Features:

- **Out-of-Band Management:** Special ports are used in isolating management traffic and out of production networks and maintain access even in the event of network problems.
- **Dense Port Density:** 48 Gigabit Ethernet ports and 4 x 10G SFP+ uplinks that can connect management working environment desks, IP phones and security devices.
- **Power over Ethernet (PoE+):** Provides up to 740W to connected equipment including IP phones, cameras and other PoE devices.

- **Redundant Architecture:** Hot-swappable power supplies and fans are modular such that they can be turned off during a failure, and satisfy Tier III concurrent maintainability.
- Ready-to-Use Management Built-in Arista EOS, CLI, SNMP, and REST API make configuration, monitoring and automation easier.
- **Operation Cost-Effective:** Less complexity and operation cost than the production-layer switches, which fits in limited IT resources.
- **Isolation of Security:** This is done physically to isolate the management traffic so that it is not interfered with by production network problems.

Justification:

Out-of-band management means having the ability to connect to any device in the IT network to ensure control in cases of failures of the networks and Cisco Catalyst 9200 meets this requirement. It offers isolation through which the management network is physically separated so that problems in the production network cannot affect management accessibility. It also has Power over Ethernet (PoE) feature that enables the operations center to directly connect to the network in order to power its management workstations, IP phones, and security cameras. It is also focused on simplicity and the switch provides built-in management features which are not as complex to setup and administer, but also the cost is kept low, which makes it an effective and reliable IT network management tool.

2.2 Routers

Routers involve the movement of data packets between two networks and identify the best paths using routing tables and protocols (Latif et al., 2015). In the case of HDS Nepal, routers will link the data centre with external networks such as internet service providers, cloud providers, as well as customer sites.

2.2.1 Selection of Core Router

Arista 7280CR2 Series Core Router

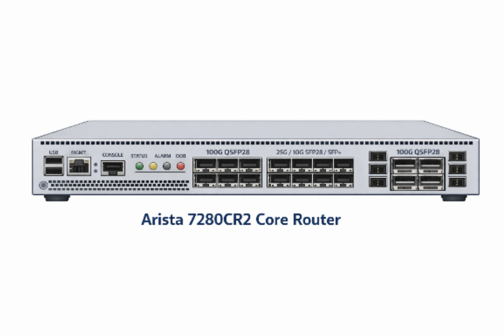


Figure 6: Arista 7280CR2 Core Router

Table 4

Feature	**Specification**
Ports	12 x 25G, 4 x 100G QSFP28
System Bandwidth	Up to 1.2 Tbps
Throughput	190 Gbps full duplex
SD-WAN	EOS integrated SD-WAN support
Security	MACsec, IPsec
Redundancy	Dual hot-swappable PSUs and fans
Management	EOS, REST API, NETCONF/YANG

Key Features:

- Multi-100G WAN routing.
- Application-aware routing SD-WAN integration.
- Traffic engineering and segment Routing of programmable networks.
- Encryption and packet processing hardware accelerators.

Real-time data monitoring. Telemetry Model-driven telemetry offers real-time visibility of routing performance.

Figure 7: Arista 7280CR2 Core Router Rear Panel

Justification:

The choice of the Arista 7280CR2 is possible in HDS Nepal as it offers decent connectivity, which facilitates high -speed WAN, cloud and MPLS VPN connections to support a wide range of network needs. The native SD-WAN functions of it match perfectly with SDN overlay architecture, allowing advanced network management and automation. The switch is a high performance switch that is able to meet the current and future requirements of the traffic and the inbuilt encryption assures that the switch will meet the strict financial security requirements. These and other redundancy features, such as dual power supplies, fans, etc., enable Tier III concurrent maintainability, available 247. Also, its programmability based on REST API and NETCONF/YANG allows automated workflows making it easier to operate even when the local networking experience is limited.

2.3 Firewalls

Firewalls are used to regulate the network traffic in accordance with a set of predefined security rules, and they represent the border between trusted internal networks and untrusted external networks (Ullrich et al., 2016). In case of HDS Nepal, the next generation firewalls (NGFW) offer to protect against advanced threats, applications visibility, and intrusions prevention. In this context, the vendor of a Next-Generation Firewall should be selected based on personal preference and the specific qualities required in a firewall solution, including but not limited to: nGFW, intrusion detection, and attacks on applications and OS.

2.3.1 Selection of Firewalls

Model: Fortinet FortiGate 1800 F.

Figure 8: Fortinet FortiGate 1800 F

Feature Specification

- Form Factor 2RU fixed platform.
- Firewall Throughput 134 Gbps
- NGFW Throughput 18 Gbps (with IPS, Application Control)
- Threat Protection 9.5 Gbps
- IPS Throughput 25 Gbps
- Multi-Sessions 20 million.
- New Sessions/Second 700,000
- Ports 18 x 10GE SFP+, 8 x 25GE SFP28, 4 x 100GE QSFP28

- VPN Throughput 36 Gbps (IPsec) and 4.5 Gbps (SSL).
- Redundancy Hot spare power supplies, two.
- Management FortiOS, FortiManager, FortiAnalyzer, REST API

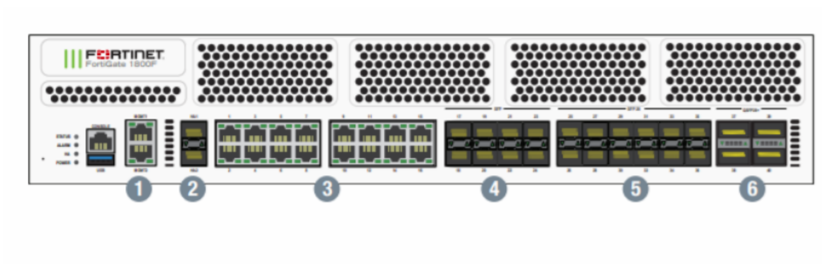


Figure 9: Fortinet FortiGate 1800F Front Panel

Key Features:

High Throughput: 134 Gbps firewall throughput of all internal and external traffic does not choke.

Detection Advanced protection: Built-in IPS, antivirus, web filters, and application control combined with AI-powered threat intelligence (FortiGuard Labs)

SSL/TLS Inspection: Hardware accelerated decryption/encryption of encrypted traffic without affecting the performance.

SD-WAN Integration: In-service SD-WANs are in compliance with the broad SDN architecture.

Automation: Security Fabric automation is the coordination of viewpoint of network device threats responses.

Multi-Tenancy: Virtual domain (VDOMs), are used to provide isolation between dissimilar client environments.

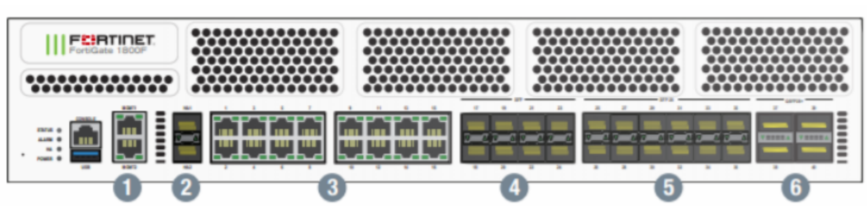


Figure 10: Fortinet FortiGate 1800F Rear Panel.

Justification:

The firewall model that is chosen as the main firewall to be used in HDS Nepal is Fortinet FortiGate 1800F because it is highly performing, scalable, secured and economical. It supports 134 Gbps firewall throughput and 18 Gbps NGFW throughput, which ensure that the important applications like real time banking transactions can be executed without latency. The firewall can have up to 20 million parallel connections enabling HDS Nepal to handle its existing customer base as well as with the anticipated 30-40 percent yearly growth. Its Virtual Domains (VDOMs) has a multi-tenant security capability that allows the logical isolation of various clients like banks, e-commerce sites and government organizations in a common infrastructure. The SD-WAN and REST API built-in also adhere to SDN overlay architecture, allowing policy provisioning to be automated, and they can be managed centrally. Threat intelligence offered by FortiGuard Labs, based on AI, is real-time protection against new threats, which is essential in the security of the financial sector.

2.3.2 Firewalls of internal segmentation.

Model: Fortinet FortiGate 600F.

Figure 11: Fortinet FortiGate 600F.

Specifications:

- Feature Specification
- Firewall Throughput 40 Gbps
- NGFW Throughput 4.4 Gbps
- Simultaneous Sessions 4 million.
- Ports 12 x GE RJ45, 12 x GE SFP, 4 x 10GE SFP+

- Form Factor 1RU
- Management FortiOS combined with FortiGate 1800F.

Justification:

Defense-in-depth is achieved by internal segmentation firewalls placed between functional areas (e.g. between the web servers, a set of application servers and a set of databases).

Micro-Segmentation: Imposes nothing short of zero-trust by limiting horizontal data centre movement.

Young compliance: Meets worldly network compliance standards that demand network segmentation.

Centralized Management:managing on the same platform as the edge firewalls that has made operations easy.

3. Server Infrastructure

A server is an operational system that is specialized to deliver services, data or resources to the client computers within a network (Monteclaro, 2023). Servers in HDS Nepal data centre support:

Core banking applications

- Database management software (Oracle, MySQL, and PostgreSQL).
- Platforms in virtualization (VMware vSphere)
- Container orchestration (Kubernetes)
- Application hosting and Web servers.
- Authentication services and directory services.
- Disaster recovery and back up services.

3.1 Types of Servers

3.1.1 Rack Servers

Standalone rack servers are also designed to be fitted on standardized racks, where they deliver equal performance, scalability, and serviceability (Nada & Elfke, 2017).

Figure 12: Rack Server as an example of Dell PowerEdge R760.

Characteristics:

- **Serviceability**: It is easily accessible, replaceable or upgradable.
- **Cooling:** Server to server fan.
- **Connectivity:** Front / rear I/O independent network ports.

Advantages:

- Self-scheduling (when one server goes down, the rest are not impacted)
- Configurations can be customized in each server.
- A similar problem easier to troubleshoot and replace.
- Broadly and broadly-vendor capable.

Disadvantages:

- Reduced density in comparison with blades (reduced number of servers per rack)
- Increased cabling (power and network per server) is needed.
- Low power efficiency per server.

Use Cases at HDS Nepal:

- High storage capacity database servers.
- Servers, with certain hardware needs, which can be of the application server type.
- Servers management and infrastructure.
- Tasks that involve physical isolation.

3.1.2 Blade Servers

Blade servers are ultra-dense, modular servers which are slid in a shared chassis which provides power, cooling, and network connectivity.

Figure 13:: Blade Server is an example of HPE Synergy 480 Gen11

Characteristics:

- Form Factor: Blade modules that do not need a chassis that can fit in multiple slots.
- Shared Resources Chassis Sources power, cooling, management modules.
- High density 8-16 blades in chassis (8U-10U)
- Chassis management controller: Centralized Management.
- Integrated Networking: Backplane Super switches and integrated switches in chassis backplane.

Advantages:

- Maximum number of servers on a rack.
- Shared power (reduced cabling) Network backplane (reduced cabling)
- Reduced power usage per server (Posted on 2010-12-02)

Centralized management

Quick deployment (automatic)

Disadvantages:

- Greater investment (chassis needed) in the start-up.
- Blades have to match the chassis Lock-in on vendors.
- Components (chassis) of failure.

- High-density configurations have difficulties with cooling.
- Small increment in storage of each blade.

Use Cases at HDS Nepal:

- VMware ESXi hosts Virtualization clusters.
- Container platforms (nodes of the Kubernetes system)
- Heavy workloads in high density computation.
- Areas that need to scale fast.

3.1.3 Tower Servers

Tower servers are isolated systems that look like desktop computers and can be applied in both small-scale and small office settings.

Of course, the figure above can be removed when space is limited.<|human|>Naturally, the figure may be dropped when there is a lack of space.

Figure 14: Tower Server Sample - Dell PowerEdge T560

Characteristics:

- Form Factor: Standalone tower (no rack-mount conversion kit)
- Self-Contained: Power supply, cooling and expansion integrated.

- Less Dense: Does not make use of racks.

Advantages:

- Lower initial cost
- Easy installation and configuration.
- Silent work (appropriate in the office settings)
- Effective growth potential.

Disadvantages:

- Poor space efficiency
- Not optimized to use data centres.
- Low density and scalability.
- Increased per-server administration.

Use Cases at HDS Nepal:

- Do not recommend use in production data centres.
- Only appropriate in a branch office or in a development/test environment.

3.2 Server Comparison

Table 5

Feature	Rack Server	Blade Server	Tower Server
Form Factor	1U/2U/4U standalone	Blade in chassis	Standalone tower
Density (per 42U Rack)	20-42 servers	50-80+ servers	5-10 servers
Initial Cost	Moderate	High (chassis required)	Low
Per-Server Cost	Moderate	Lower (shared resources)	Higher (per-server overhead)
Power Efficiency	Moderate	High (shared PSUs)	Low
Cabling Complexity	High (per-server cabling)	Low (chassis backplane)	Moderate
Scalability	Good (add servers)	Excellent (add blades)	Poor
Serviceability	Easy (individual)	Moderate (chassis-dependent)	Easy
Vendor Flexibility	High (mix vendors)	Low (single vendor/chassis)	High
Management	Per-server or cluster	Centralized (chassis)	Per-server
Cooling Requirements	Moderate	High-density challenges	Low
Typical Workloads	General purpose, databases	Virtualization, HPC	Small business, branch

3.3 Server Solution Recommended.

The hybrid rack and blade server strategy is suggested based on the needs of HDS Nepal regarding virtualization, database performance, scalability, and operational efficiency.

Main Compute Server: HPE ProLiant DL380 Gen11 (Rack Server)

Figure 15: HPE Proliant DL380 Gen11 Server.

Specifications:

Feature Specification

Form Factor 2U Rack

CPU 2 Intel Xeon Scalable 5th Gen (totals up to 64 cores)

Memory Up to 8TB DDR5 (32 x 256GB) @ 5600 MT/s

Storage 30 SFF NVMe/SATA drives

|human|>Storage Up to 30 SFF NVMe/SATA drives.

Storage Controller HPE Smart Array Gen11 (tri-mode)

Platforms Networking 4 x 1/10/25GbE embedded (OCP slot expandable)

Expansion Slots Eight PCIe Gen5 Slots.

Power Supplies Dual (N+1 redundancy) hot-plug.

Management HPE iLO 6 (integrated Lights-Out)

Security Secure Recovery, Silicon root of trust.

GPU Support Up to 3 double-width GPUs (when used as an AI/ML workload)

Key Features:

- **Gen11 Platform:** Intel Scalable processors based on the latest DDR5 memory innovation and PCIe Gen5.
- **Remote Management:** iLO 6 will allow every out-of-band management, satisfying the skills gap of Nepal.
- **Redundancy:** Support of Tier III concurrent maintainability of power and fans is supported through hot-plug power and fans.

HPE Synergy 480 Gen11 (Blade Server) is a high-density compute platform.

Specifications:

Feature Specification

- Form Factor Half-depth blade (fits HPE Synergy Frame)
- Processor 2 x Intel Xeon Scalable 5th Gen.
- Memory Up to 4TB DDR5 (16 x 256GB)
- Storage 2 x SFF NVMe/SATA (mezzanine expandable)
- Networking Built 10/25GbE (through Synergy Virtual Connect)
- Management HPE OneView (built into chassis)
- Production Support HPE Synergy 12000 Frame (12 blades per 10U)

Figure 16: HPE Synergy 12000 Frame.

Key Features:

- **Ultra-Density:** 12 blades per 10U chassis (more than 40 blades/ per rack)
- **Composable Infrastructure:** Resources are composable on-demand around particular workloads.
- **Single API:** HPE OneView is a solution that offers compute, storage and networking infrastructure automation.
- **Virtual Connect:** Network profiles trail blades in reprovisioning.
- **Redundancy:** power, cooling N+1 in chassis.

3.3.1 Reason why Hybrid Server should be selected.

The hybrid server implementation that involves the use of HPE ProLiant DL380 Gen11 rack servers and HPE Synergy 480 Gen11 blade servers is chosen to support various workloads in the HDS Nepal data center. Rack servers offer great storage capacity and PCIe expansion to database and application workload, whereas the blade servers offer high-density compute power that is used in virtualization platforms like VMware ESXi. This design enhances the performance by isolating I/O-intensive and CPU-intensive processes and it can be expanded as demand increases. HPE iLO 6 and HPE OneView are centralized so that operation becomes less complex and allows the administration to be conducted remotely. Moreover, unnecessary elements, power-saving processors, and scalability to the future technologies are guaranteed to achieve Tier III reliability, decrease energy usage, and extendability over time to the data center.

4. Storage Systems

A data centre storage system is specialized infrastructure that is used to store, administer, and defend computer-generated data to support applications programs, services, and users (Qin et al., 2018). In the case of HDS Nepal, there are storage requirements:

Financial database-high-performance (low latency, high IOPS) storage.

Storage capacity of each backups and archives.

Joint storage (datastores of VMware virtualization).

Cloud-native application object storage.

Replication of a disaster recovery.

4.1 Types of Storage Systems

4.1.1 Direct-Attached Storage (DAS)

DAS is employed in high throughput and low latency networks that do not need a network intermediary and shares are limited.

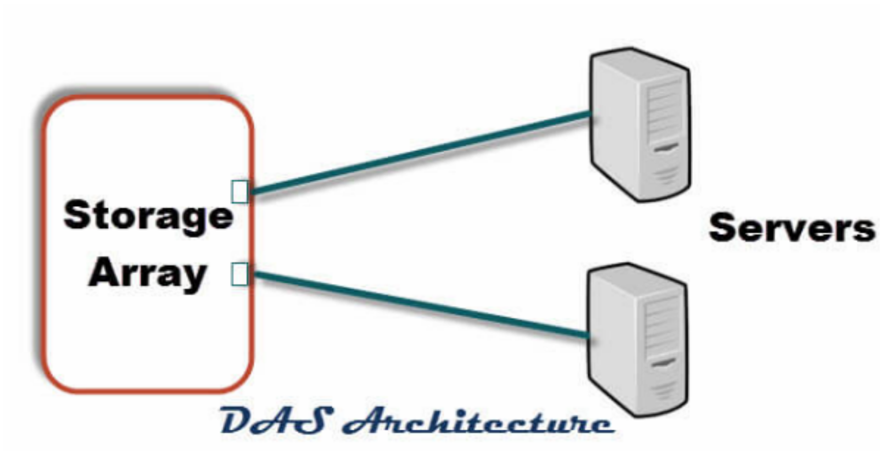


Figure 17: DAS Architecture Diagram

Characteristics:

Connection Direct server connection (SAS, SATA, NVMe)

Storage: Server OS is a storage manager.

Scalability: Limit on chassis capacity of servers.

Advantages:

If the source and epibots are in one rack, this is the shortest latency (direct connection).

Simple implementation

Reduced price when computing one server.

No network storage traffic

Disadvantages:

- No sharing amid servers except involving complicated configurations.
- Wasted capacity (allocated to individual servers)
- Limited scalability
- Backup issues (have to be backed up via the server)

Use Cases at HDS Nepal:

- Boot drives in the operating systems.
- Local temporary storage

4.1.2 Network Attached Storage (NAS)

NAS offers file level access to the storage via a standard IP network, which is optimal in sharing files and user data storage.

Characteristics:

- Protocol: NFS, SMB/CIFS
- Access: File-level
- Network: 1 /10 /25 GbE as a standard Ethernet.
- Dedicated NAS appliance Management: NAS-ID Software.

Advantages:

- Basic file sharing between a series of servers/clients.
- Easy management
- User data and file services are good.
- Windows compatibility, Protocol compatibility, Linux compatibility, MacOS compatibility

Disadvantages:

- The scalability of the file system is limited.
- Enterprise is dependent on network conditions.

Use Cases at HDS Nepal:

- User home directories
- Shared file repositories
- Web content storage
- Backup targets (secondary)

4.1.3 Network Attached storage (NAS).

NAS allows access to storage at file level via a standard IP which is good when there is need to share files among others as well as user data.

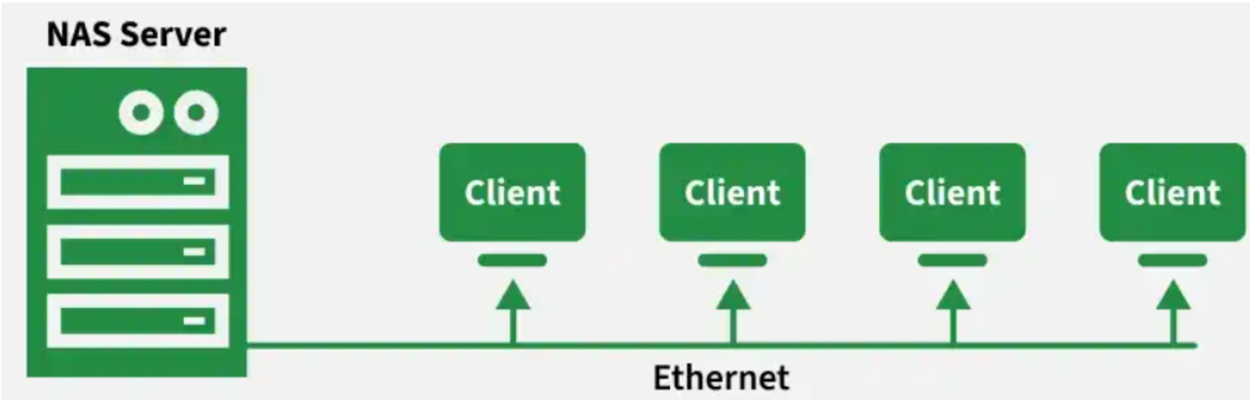


Figure 18: NAS Architecture Diagram

Characteristics:

- Protocol: NFS, SMB/CIFS
- Access: File-level
- Connection: Standard Ethernet (1/10/25GbE)
- Management: NAS appliance application.

Advantages:

- Basic file sharing between three or more servers/clients.
- Easy management
- User data and file services Good.
- Linux, MacOS compatibility (Windows, Linux, macOS)

Disadvantages:

- Slower than block storage File level overhead.
- Inappropriate with a database workload.
- File system is limited due to scalability.
- The performance is conditional on the network conditions.

Use Cases at HDS Nepal:

- User home directories
- Shared file repositories
- Web content storage
- Backup targets (secondary)

4.1.4 Storage Area Network (SAN)

SAN is a block level storage that is accessed via a special high speed network and is designed to support performance sensitive applications such as databases and virtual machines.

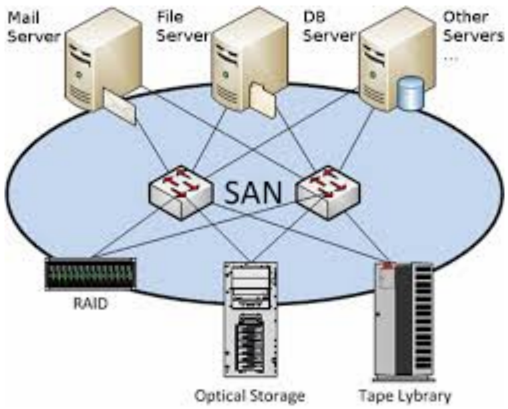


Figure 19: SAN Architecture Diagram

Characteristics:

- Protocol: Fibre channel, iSCSI, NVMe-oF.
- Access: Block-level (LUNs)
- Network: Dedicated Fibre Channel (storage) or converged (Ethernet).
- Management: Centralized Managing storage arrays.

Advantages:

- Maximum performance (minimum latency, maximum IOPS)
- Cross server shared storage.
- High-tech (snapshots, replication, thin provisioning)
- The management and capacity utilization are centralized.
- Virtualization datastores (VMFS).

Disadvantages:

- Higher complexity
- Greater price (dedicated fabric, enterprise arrays)
- Specialty skills are required.

Use Cases at HDS Nepal:

- Storage (database, Oracle, MySQL).
- Virtually machine datastores (VMware).

- Mission-critical applications
- Primary production storage

4.2 Storage System Comparison

Feature DAS NAS SAN

- Access Type Block File Block
- Direct (SAS/SATA) Network Direct (Fibre channel, iSCSI, NVMe-oF) Direct (SAS/SATA) Network
- Latency < 100 µs 1-5 ms < 500 µs (FC/NVMe-oF)
- IOPS Very high (local NVMe) Moderate Very high (enterprise arrays)
- Sharing Single server Multiple servers/clients Multiple servers.
- Scalability Server-limited Good (scale-out NAS) Excellent (scale-up/out)
- Management Per-server Centralized Centralized.
- Advanced Features Limited Snapshots, replication, Snapshots, replication, thin provisioning, deduplication, compression.
- Cost/TB Low Moderate Higher
- Complexity N Low N Moderate N High.

Boots and temp storage Use Case Boots and temp storage Use Case File services, backups Data Use Case databases, virtualization, production Use Case

4.3 Recommended Storage Solution: Storage Area Network (SAN)

Main Storage Array: Dell PowerStore 500T.

Specifications:

Form Factor 2U base enclosure

Type Storage All-NVMe (end-to-end NVMe architecture)

Maximum Capacity Maximum of 1.2 PB per appliance (raw)

Peak IOPS1.2 million (random read 4KB)

Latency < 1 ms (typical)

Controllers Dual active-active (fully redundant)

Host Interconnectivity 32Gb Fibre Channel, 25Gb iSCSI, NVMe-oF.

Supported drives NVMe SFF (2.5")- supported mixed drives.

Data Reduction 4:1 (usual) inline deduplication and compression.

High-end Features Snapshots, meta replication, metro replication, native async replication.

Management PowerStore Manager, REST API, Ansible module.

Integration VMware vSphere integration (VAAI, VVols)

Key Features:

- End-to-End NVMe: Drives to host direct connectivity, over the top of old-fashioned SAS/SATA emaciations.
- Active-Active Controllers: The two controllers are operating at the same time, which makes the controllers as productive as they can be.
- Inline Data Reduction: Hardware-based deduplication and compression lowers the actual storage expenses.
- Anytime Upgrade: Non-disruptive controller upgrades.
- AppsON: Built-in option to operate said applications on the array.
- Metro Replication Synchronous replication of zero RPO disaster recovery.
- Cloud Integration: Public cloud tiering and public cloud backup.

Backup/ Archive (Secondary Storage): Dell EMC Data Domain DD3300.

Specifications:

Feature Specification

Form Factor 2U appliance

Maximum Capacity Maximum 144 TB (maximum deduped logical capacity: maximum 720 TB)

Deduplication Ratio 10:1 -30:1 normal.

Protocols NFS, CIFS, VTL, OST, DD Boost.

Throughput Up to 5.5 TB/hour

Replication Data Domain Replicator (encrypted)

Integration Veeam, Commvault, NetBackup, and so on.

Justification:

- Data Domain appliance offers:
- High Deduplication: Cuts backup storage usages by 10-30x which brings down the cost to a great extent.
- Ransomware Protection: Retention lock feature denies backup a chance to be modified/ deleted.
- Disaster Recovery: Second site replication.
- Veeam integration Veeam Backup and Replication is a native integration of the Veeam Backup and Replication software (Veeam backup software used by HDS Nepal).

4.4 Reason behind SAN Selection.

The Dell PowerStore 500T SAN will be used as the major storage solution of HDS Nepal on the basis of:

1. Performance Requirements:

- 1.2 million IOPS and less than 1 ms latency satisfies the workloads at the financial databases (Oracle, MySQL) with high rates of transactions.
- All-NVMe architecture does away with storage bottlenecks of virtualized architectures.
- Active-active controllers have guaranteed that there is no performance degradation in event of a controller failure.

2. Virtualization Integration:

- Integrated VMware vSphere Native VMware vSphere (VAAI with offloaded operations, VVols with granular policy-based controls)
- Snapshot and replication at the VM level makes data protection easier.
- Endorses the VMware virtualization plan of HDS Nepal.

3. Scalability:

- The current requirements can be expanded by up to 1.2 PB per appliance.
- Non-disruptive scalability provides the capacity to be added without down time.
- Finances the 30-40% growth per annum of HDS Nepal.

4. Data Efficiency:

- The effect Can store 4:1 effective data at reduced costs.
- Real time deduplication and compression at no cost.
- Reduced TB power and cooling needs.

5. Tier III Compliance:

- Unnecessary controllers, power, fans.
- Invisible software updates and replacing parts.
- Metro replication assists in the need to recover disasters.

6. Management Efficiency:

- REST API allows automation, which is less burdensome.
- PowerStore Manager has easy to use GUI to perform regular tasks.
- Cope with the proud workforce shortages in Nepal using automation.

7. Multi-Protocol Support:

- Database server fibre channel with low latency.
- iSCSI of cheap server connectivity.
- NVMe-oF in the future: performance needs.

8. Disaster Recovery:

- Native asynchronous copying to the secondary site.
- Metro replication with zero RPA needs (expansion in the future)
- Reasoning complies with the business continuity goals in Section A.

Storage Tiering Strategy:

Tier Storage Type array Workloads.

Tier 0 (Performance) NVMe SSD PowerStore 500T Databases, high-transactions VMs.

Capacity (Tier 1) SSD (power expansion) PowerStore 500T General-purpose VMs

Tier 2 (Backup) High-density HDD Data Domain DD3300 Backups, archives.

Tier 3 (Object) Object storage (future) Dell ECS Object cloud-native applications, long-term retention.

5. Mapping Network Layer Component.

The table below associates the suggested IT hardware with the network layers in the Spine-Leaf architecture of SDN Overlay in Section A.

Table 6

Layer	Component	Model	Quantity (Phase 1)	Purpose
Spine Layer	Spine Switch	Cisco Nexus 9508	2	Core backbone, interconnects all leaf switches
Leaf Layer	Leaf Switch (ToR)	Cisco Nexus 93180YC-FX3	20 (1 per rack)	Server connectivity, VXLAN termination
Core Routing	Core Router	Cisco Catalyst 8500-12X4QC	2	WAN connectivity, SD-WAN, internet peering
Edge Security	Next-Generation Firewall	Fortinet FortiGate 1800F	2 (Active-Active)	Perimeter security, threat prevention
Internal Security	Internal Firewall	Fortinet FortiGate 600F	4	Zone segmentation, micro-segmentation
Management Network	Management Switch	Cisco Catalyst 9200-48P	2	Out-of-band management network
Compute - Virtualization	Blade Server	HPE Synergy 480 Gen11	36 (3 chassis)	VMware ESXi hosts, container platforms
Compute - Database	Rack Server	HPE ProLiant DL380 Gen11	24	Database servers, application servers

Compute - Management	Rack Server	HPE ProLiant DL380 Gen11	6	Management infrastructure
Primary Storage	SAN Array	Dell PowerStore 500T	2 (Active-Active)	Production storage
Backup Storage	Backup Appliance	Dell Data Domain DD3300	1	Backup and archive storage
Storage Networking	Fibre Channel Switch	Brocade G620 (optional)	2	FC SAN fabric (if FC required)

Redundancy Configuration:

- Leaf Switches Each server is linked to twoleaf switches (server NIC teaming).
- Routers: 2 routers BGP into a variety of ISPs.
- Storage: PowerStore 2 replication active arrays.

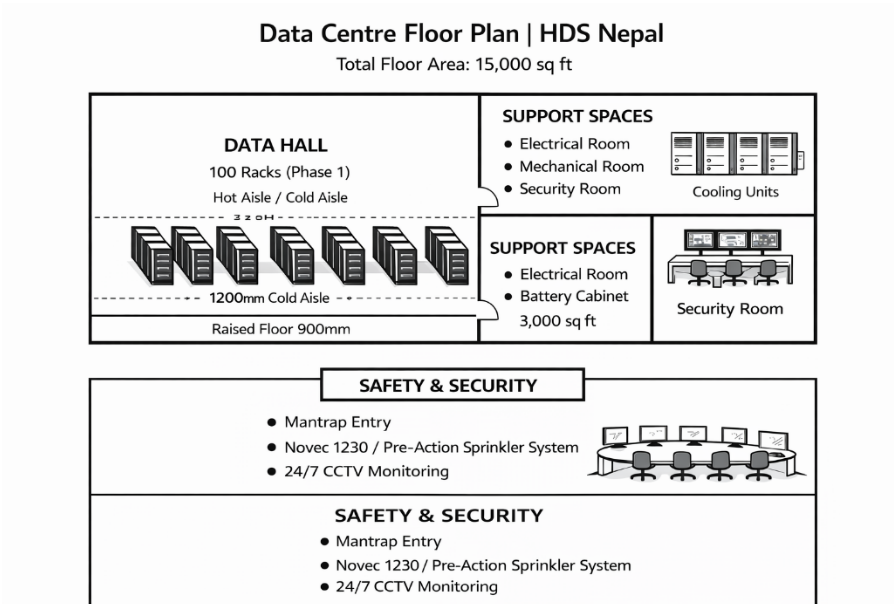
5. TIA-942 Standard IT Space Design

The TIA-942 standard is used to specify best practices in designing a telecommunications infrastructure in a data centre in order to guarantee reliability, performance and scalability. In the case of Tier III facility of HDS Nepal, TIA-942 compliance will guarantee:

- Functional areas to have unified operations.
- Maintaining cabling Structured cabling
- Energy efficient airflow is optimized.
- Zonal security obstruction.

5.1 Proposed Data Centre Layout

Table 7: Data Centre Floor plan HDS



Facility Specifications:

Table 8

Parameter	**Specification**
Total Floor Area	15,000 sq ft
Data Hall Area	10,000 sq ft
Support Spaces	3,000 sq ft (electrical, mechanical, security)
Office/Operations	2,000 sq ft
Rack Count (Phase 1)	100 racks (scalable to 200+)
Rack Dimensions	600mm x 1200mm (standard)
Aisle Configuration	Hot aisle/cold aisle (1200mm cold, 1000mm hot)
Floor Loading	1200 kg/m² (raised floor: 900mm)
Fire Suppression	Pre-action sprinkler + inert gas (Novec 1230)
Physical Security	Mantrap, biometric access, 24/7 CCTV

5.2 Zone Descriptions

5.2.1 Entrance Room (ER)

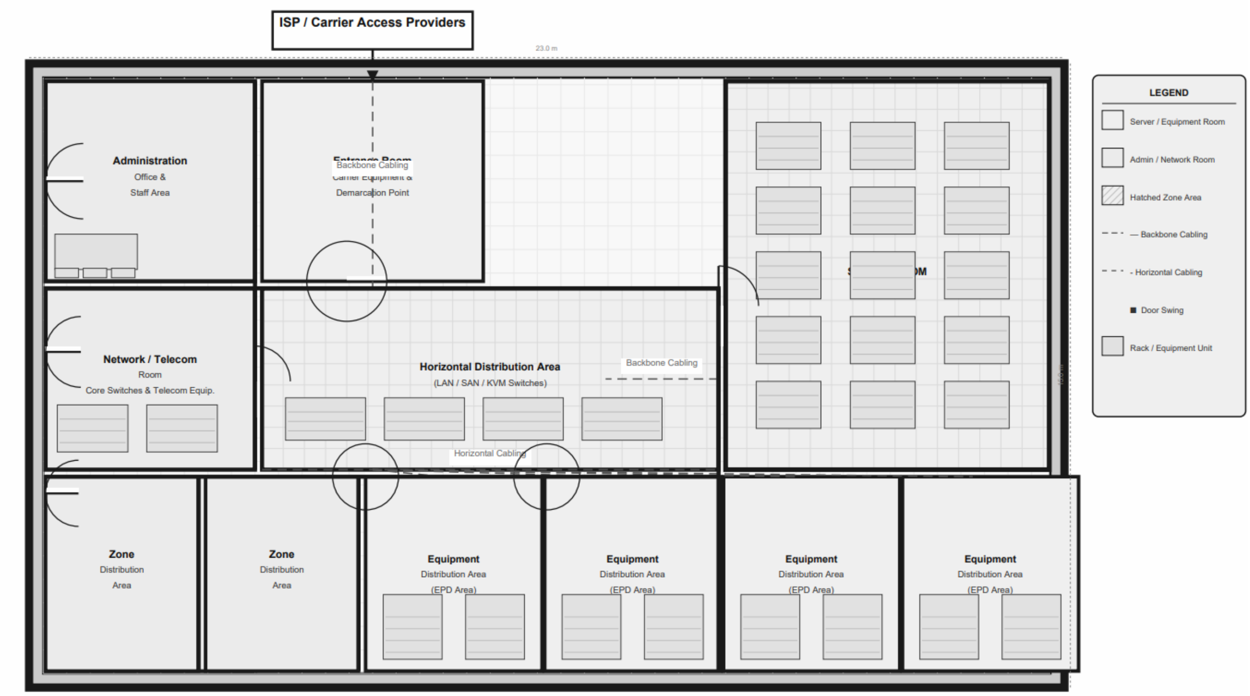


Figure 20: Entrance Room Layout

- Position: Data hall: next to loading dock.
- Function: Cabling (telecom, ISP, cloud providers) termination point.

Components:

- Demarcation points of the service providers.
- Main distribution frame (MDF)
- Optical distribution frames (ODF).

Design Considerations:

- Physically distanced with data hall to secure it.
- Access control security zone.
- Eliminate redundant carrier diversity access routes.

TIA-942 Conformity: ER should be available without having to enter data hall, and with secured cableways to MDA.

5.2.2 Main Distribution Area (MDA)

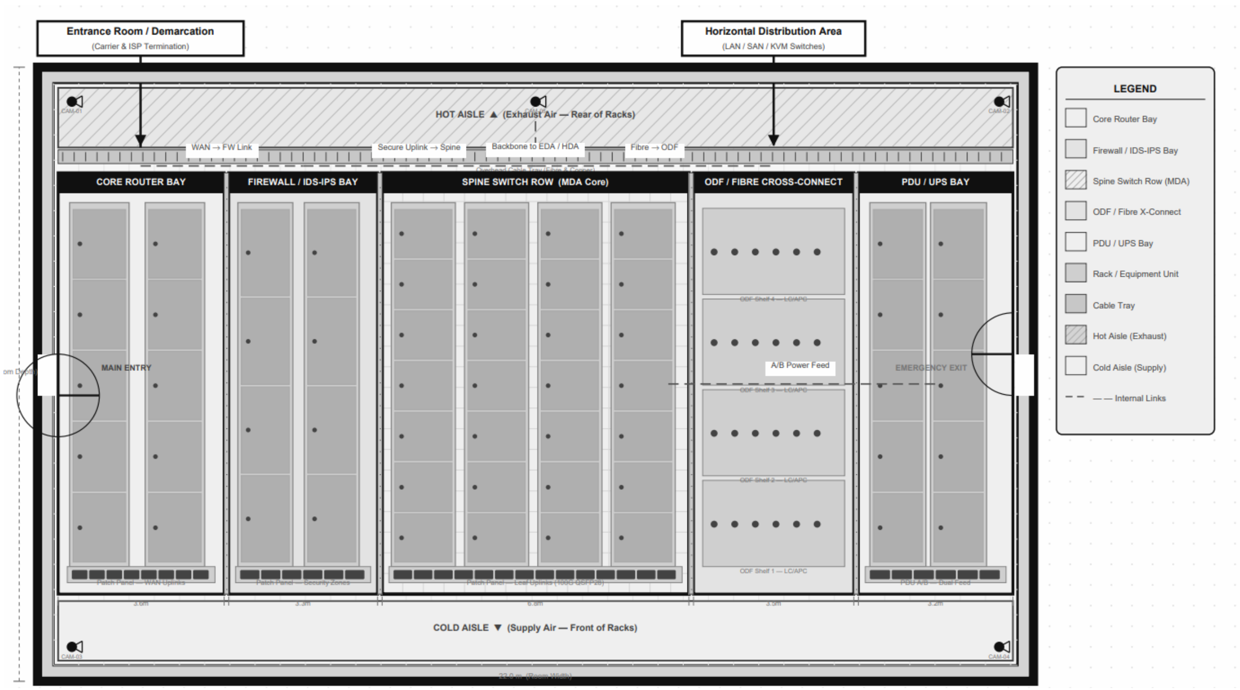


Figure 21: Main Distribution Area Layout.

Location: Data hall central location.

Application: Central location of network and storage equipment.

Components:

- Core switches (Cisco Nexus 9508) - 2 racks.
- Core routers (2 racks Cisco Catalyst 8500)
- Firewalls (FortiGate 1800F) - 2 racks
- SAN directors/ F fibre switches - 2 racks.
- SDN controllers(physical/virtues)

Design Considerations:

- Directly linked to all racks of HDA through structured cabling.
- Top level security area in data hall.

- Specialized cooling areas (increased density).
- Dual supporter fed on independent PDUs.

TIA-942 Compliance: MDA is a central point to all the horizontal cabling to HDAs.

5.2.3 Horizontal Distribution Area (HDA)

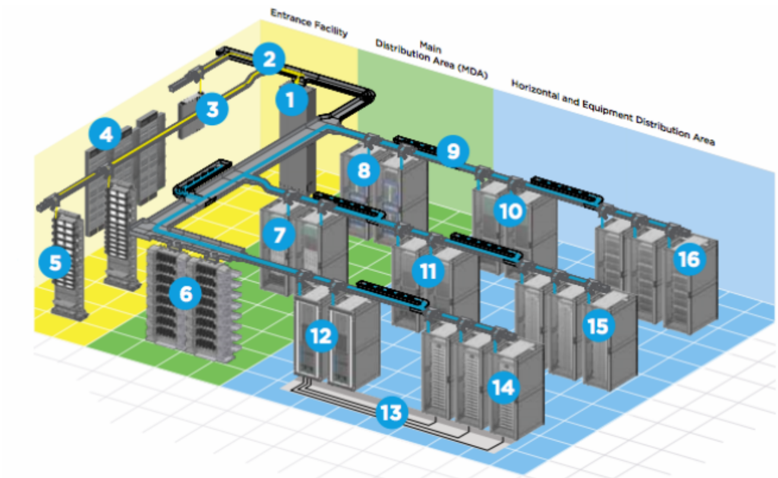


Figure 22: Horizontal Distribution Area Layout.

Location: At the end of every row of equipment racks.

Purpose: Between MDA and individual equipments.

Components:

- Organized points of termination of cabling.
- Row-level PDU distribution
- Cable (Vertical and horizontal) management.

Design Considerations:

- Monolithic or mono-block HDA.
- The interpretation of Iceland's internet adoption remains immature.<|human|>Senseless linkage to both MDA switches.
- Zone labeling per ANSI/TIA-606-B
- Overhead- Pathway cable troughs.

TIA-942 Compliance: The TIA-942 establishes compliance through connection between MDA and EDA via structured cabling.

5.2.4 Zone Distribution Area (ZDA)

Figure 30: Zone Distribution Area Layout

Location: Free point of consolidation between HDA and EDA.

Function: Allows the reconfiguration flexibility without interrupting HDA.

Components:

- Passive patch panels
- Cable management

Design Considerations:

- Applied in dynamic systems where there is need of frequent movement/add/change.
- Placed in convenient locations (above or at the floor).
- Stamped with tags on them.

TIA-942 Compliance: Non-compulsory, however, advised in flexible environments.

5.2.5 Equipment Distribution Area (EDA).

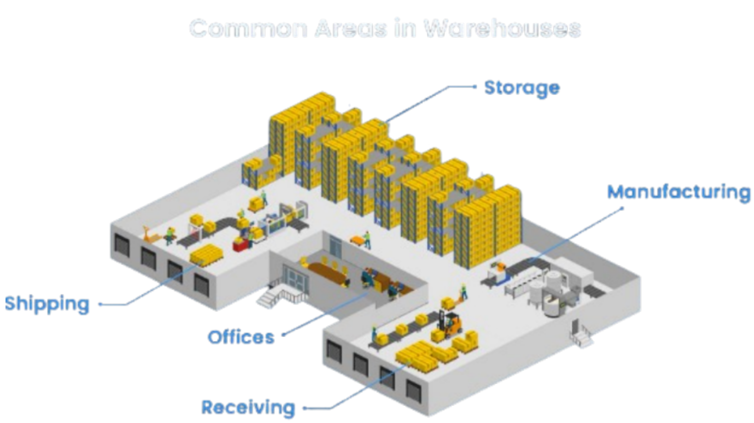


Figure 23: Equipment Distribution Area Layout.

Location: Rack of individual equipment in each data hall.

Function: where IT equipment (servers, storage, switches, etc.) is physically located.

Components:

- Equipment racks (42U)
- Servers (rack and blade)
- Storage arrays
- ToR switches (leaf switches)
- Intelligent PDUs (rack-level)
- KVM switches (if required)

Design Considerations:

- Dual supporter fed on independent PDUs.
- Identical network links to two leaf switches.
- Server cabling cable management weapons.
- RFID tracking and asset tagging.

TIA-942 Compliance: EDA cabling is based on structured cabling, and maximum distances are stipulated.

5.3 Detailed Zone Layout

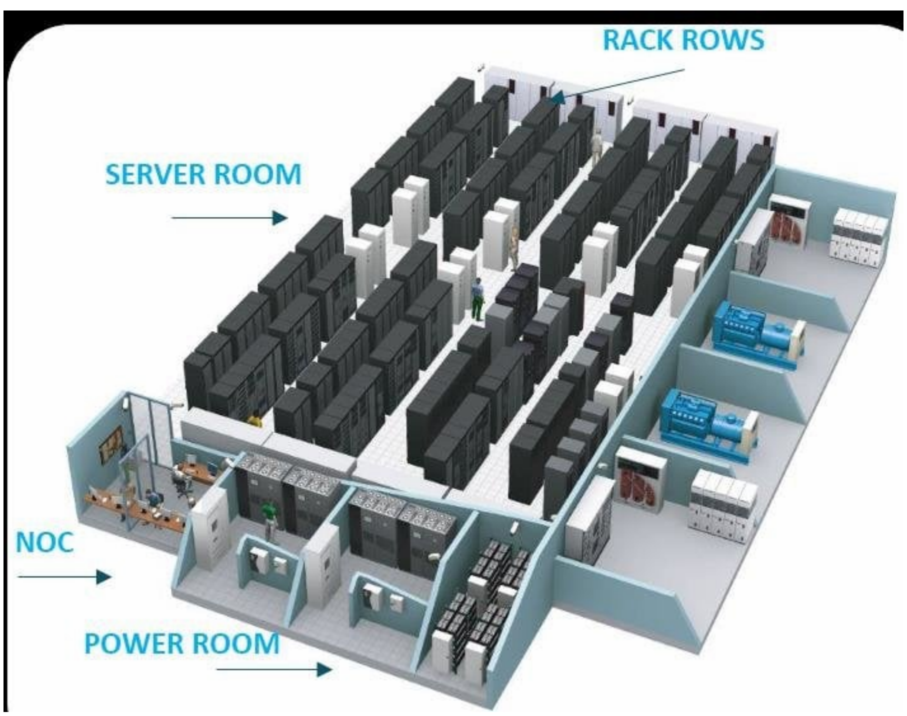


Figure 24: Full TIA-942 Zone Layout and all Zones

Row Configuration (Sample Row):

Table 9

Position	**Equipment Type**	**Redundancy**
Rack 1	Leaf Switch (ToR) + Patch Panels	N+1 (redundant switches per row)
Rack 2–5	Compute - Rack Servers (Database)	Dual power, dual network
Rack 6–10	Compute - Blade Enclosures	Dual power, chassis redundant
Rack 11	Storage - SAN Array	Active-active controllers
Rack 12	Storage - Expansion Shelves	Redundant paths
Rack 13–14	Network - Distribution (if not ToR)	Redundant switches

6.4 Cabling Infrastructure

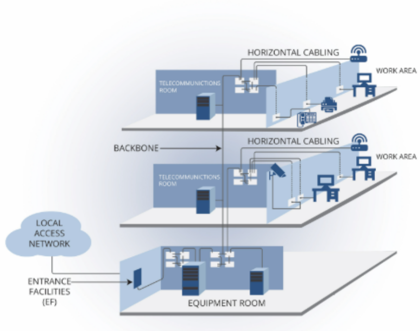


Figure 25: Structured Cabling Diagram

Cabling Standards:

Table 10

Cable Type	Application	Standard
OM4 Multimode Fiber	10/40/100G backbone (≤400m)	TIA-568.3-D
OS2 Singlemode Fiber	Long-distance, future 400G	TIA-568.3-D
Category 6A UTP	10GBASE-T (≤100m)	TIA-568.2-D

Cabling Topology:

- Leaf to Server: Category 6A (copper) of 10/25GBASE-T cable or DAC cable to make ToR connections.
- Storage: FC/NVMe-oF fiber Storage: iSCSI Category 6A.
- Management: out-of-band network: category 6A.

Cable Management:

- Fiber and copper overhead cable tray (ladder type).
- Rack to rack vertical cable managers (6" wide).
- Bend radius (protection 10x cable diameter (copper) -15x fiber)

5.4 Physical Security Zones

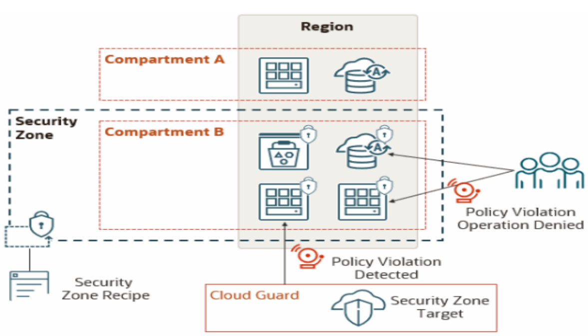


Figure 26: Security Zone Diagram.

Security Layers:

- Building Entry: Biometrical encountered mantrap, security desk.
- Hall Entry Data: Biometric + card Interlock, mantrap.
- Zone Separation: Inter zonal doors (MDA, HDA, etc.).
- Cabinet Security: user racks locks are electronically keyed.

Access Control Points:

- ER: Only the facilities and telecom persons authorized.
- MDA: Senior engineers (top restriction) only.
- Support Areas: Escort Services Maintenance staff as needed.

5.5 Environmental Monitoring

Sensors per Zone:

- Temperature (ambient temperature, rear of rack, front of rack, etc.)
- Humidity
- Differential (diffuse pressure) (containment) Air pressure (differential).
- Underfloor, cooling unit near, water leakages.

Integration: The sensors are connected to DCIM platform (Schneider Electric EcoStruxure IT), where they are centrally monitored and alerted.

6. Conclusion

Section B includes the specific implementation plan of Tier III data center of the Himalayan Digital Services Pvt. Ltd., where the strategic design proposed in Section A is converted into particular IT hardware that is available in the market, and a physical layout that is in line with the TIA-942 specifications. The hardware choice is on high performance based on the Cisco Nexus spineleaf switches and Dell PowerStore SAN with 1.2 million IOPS and HPE Gen11 servers with DDR5 and PCIe Gen5 to support the financial applications in the enterprise. Modular components will be used to assure scalability; it will be able to handle the estimated 30 to 40 percent growth of HDS Nepal without redesign. Tier III concurrent maintainability is ensured by redundant and hot swappable components, which have 99.982% uptime.

The proposed floor plan is based on the TIA-942 rules and has well-identified areas including ER, MDA, HDA, ZDA, and EDA so that the activities can be organized properly and the infrastructure can be managed effectively. The practice of standardized cabling is also a way of enhancing maintainability, and the multi-layer security between the perimeter and the cabinets of the server enhance the physical security of the infrastructure tackling the issues of seismic risks, unreliable power supply and scarcity of skilled manpower in Nepal. The entire implementation plan will bring the HDS Nepal to Tier III certification and provide highly secured and mission-critical services at 99.982 percent availability.

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